

AKHISAR, Turkey

WMO: 171840

Lat: 38.9114N

Lon: 27.8230E

Elev: 93

StdP: 100.21

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -2.8	(c) -1.4	(d) -6.3	(e) 2.2	(f) -0.3	(g) -4.8	(h) 2.5	(i) 0.6	(j) 8.3	(k) 6.7	(l) 7.0	(m) 6.4	(n) 0.9	(o) 350	(p) 0.405

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.8	38.3	21.6	36.7	21.5	35.2	21.3	24.0	34.4	23.1	33.2	22.4	32.3	2.5	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.7	15.6	29.6	19.8	14.7	28.4	18.9	13.9	27.6	72.5	34.5	69.0	33.1	66.1	32.3	27.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 7.1	(b) 6.0	(c) 5.2	(e) -5.6	(f) 41.2	(g) 2.0	(h) 1.6	(i) -7.0	(j) 42.3	(k) -8.2	(l) 43.2	(m) -9.3	(n) 44.1	(o) -10.7	(p) 45.2		
(a) 7.1	(b) 6.0	(c) 5.2	(e) -6.2	(f) 25.2	(g) 2.0	(h) 1.4	(i) -7.7	(j) 26.2	(k) -8.8	(l) 27.0	(m) -10.0	(n) 27.8	(o) -11.4	(p) 28.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	17.1	6.5	8.2	10.8	15.2	20.5	25.2	28.1	28.1	23.6	18.2	12.1	7.7		
	DBStd	8.35	3.65	3.76	3.52	3.59	3.05	2.92	2.07	2.07	2.92	3.24	3.68	3.79		
	HDD10.0	339	118	71	33	4	0	0	0	0	0	1	21	92		
	HDD18.3	1565	367	283	233	107	14	1	0	0	1	42	189	328		
	CDD10.0	2921	11	21	59	159	326	457	561	560	409	255	82	22		
	CDD18.3	1103	0	0	1	11	82	207	302	302	160	38	1	0		
	CDH23.3	12897	0	1	11	189	944	2414	3791	3617	1602	317	12	0		
	CDH26.7	6594	0	0	1	51	359	1239	2140	1995	732	77	1	0		
(14) Wind	WSAvg	1.7	1.5	1.7	1.9	1.6	1.6	2.0	2.6	2.5	1.7	1.5	1.1	1.3		
(15) Precipitation	PrecAvg	567	80	82	66	53	39	15	5	5	20	43	69	90		
	PrecMax	792	197	202	145	136	207	47	24	24	69	163	182	219		
	PrecMin	323	2	11	17	11	3	0	0	0	0	2	0	0		
	PrecStd	126	52	52	38	30	40	16	6	6	21	37	44	56		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.4	20.9	24.6	31.3	34.7	39.6	41.2	39.5	37.8	31.5	25.1	19.8		
		MCWB	12.7	13.6	15.0	16.8	19.1	21.8	22.5	21.8	20.4	19.3	16.6	14.0		
	2%	DB	16.3	18.4	21.9	28.0	32.6	36.7	38.5	37.9	34.9	28.7	22.3	17.4		
		MCWB	11.8	12.6	13.4	16.7	18.9	21.2	21.8	21.7	20.3	18.1	15.5	13.4		
	5%	DB	14.3	16.7	20.0	25.3	30.6	34.7	36.7	36.3	32.8	26.7	20.4	15.4		
		MCWB	10.7	11.6	12.8	15.8	18.5	20.8	21.5	21.7	19.7	17.5	14.5	12.3		
	10%	DB	12.3	14.7	17.9	23.0	28.6	32.9	35.1	34.8	31.0	24.8	18.4	13.7		
		MCWB	9.7	10.6	12.1	14.9	17.9	20.2	21.5	21.8	19.3	16.9	13.9	11.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.5	14.2	16.1	18.8	21.2	23.6	25.2	25.4	22.6	20.8	17.7	14.8		
		MCDB	17.2	19.1	22.9	27.7	31.0	35.8	36.1	35.6	33.5	28.1	22.9	18.1		
	2%	WB	12.1	13.0	14.3	17.4	20.2	22.4	24.1	24.2	21.6	19.4	16.2	13.6		
		MCDB	15.8	17.7	20.0	26.1	29.9	33.7	34.9	33.9	31.6	25.7	20.6	16.6		
	5%	WB	10.9	11.9	13.3	16.3	19.2	21.6	23.1	23.3	20.8	18.5	15.2	12.6		
		MCDB	13.7	15.9	18.6	24.0	28.4	32.2	33.4	32.8	30.1	24.6	19.2	15.2		
	10%	WB	9.9	11.0	12.4	15.2	18.4	20.9	22.3	22.5	20.1	17.7	14.2	11.5		
		MCDB	12.1	14.4	17.2	21.9	26.9	31.0	32.3	31.9	28.7	23.4	17.8	13.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.3	9.4	11.0	12.6	13.7	14.0	13.8	13.7	13.7	11.9	10.6	8.3		
		MCDBR	10.5	13.0	15.3	17.1	17.5	16.7	16.8	16.0	16.7	14.9	13.6	9.4		
		MCWBR	6.8	7.8	8.0	7.9	6.9	5.8	5.1	4.5	5.7	6.2	7.4	6.0		
		MCDBR	9.7	11.6	13.2	15.5	15.0	14.8	14.1	13.4	14.2	12.4	11.6	8.8		
	5% WB	MCWBR	6.5	7.2	7.4	7.6	6.4	5.4	5.0	4.6	5.4	5.8	6.8	5.9		
(40) Clear-Sky Solar Irradiance	taub		0.356	0.392	0.433	0.474	0.449	0.413	0.407	0.414	0.403	0.398	0.371	0.358		
	taud		2.354	2.224	2.108	2.026	2.164	2.304	2.320	2.292	2.311	2.309	2.342	2.363		
	Ebn at Noon		802	815	818	807	837	866	868	853	839	800	776	768		
	Edh at Noon		91	117	145	168	149	130	127	127	117	106	91	84		
(44) All-Sky Solar Radiation	RadAvg		1.96	2.66	4.00	5.26	6.48	7.38	7.77	6.86	5.34	3.70	2.49	1.80		
	RadStd		0.25	0.32	0.34	0.53	0.40	0.50	0.20	0.25	0.34	0.25	0.26	0.27		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only		N/A	N/A	N/A	+0.72	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (15 neighbors)		+0.40	N/A	N/A	N/A	N/A	N/A	N/A	-92	N/A	N/A

Nomenclature: See separate page